



PROBABILITY AND STATISTICS

QUESTION BANK

MODULE 2

SHORT QUESTION AND ANSWER

1. What is Correlation?

Answer:

Correlation is a statistical technique used to measure the degree and direction of relationship between two variables. It shows whether the variables move together, move in opposite directions, or have no relationship at all.

2.State the Range of Correlation Coefficient.

Answer:

The value of the correlation coefficient always lies between **-1 and +1**. A value of **+1** indicates perfect positive correlation, **-1** indicates perfect negative correlation, and **0** indicates no correlation.

3.What is Regression?

Answer:

Regression is a statistical method used to study the relationship between two or more variables. It helps in estimating or predicting the value of one variable based on the known value of another variable.

4.Write the formula for Karl Pearson's correlation coefficient.

Answer:

$$r = \Sigma xy / \sqrt{(\Sigma x^2 \Sigma y^2)}$$



5. What are Regression Lines?

Answer:

Regression lines are lines of best fit that represent the relationship between two variables. There are two regression lines: the regression line of **Y on X** and the regression line of **X on Y**.

6. If $b_{xy} = 0.25$ and $b_{yx} = 0.64$, find r .

Solution:

$$\begin{aligned} r &= \sqrt{b_{xy} \times b_{yx}} \\ r &= \sqrt{0.25 \times 0.64} \\ r &= \sqrt{0.16} = 0.4 \end{aligned}$$

Answer: $r = 0.4$

7. Define Rank correlation

Rank correlation measures the strength and direction of the association between two ranked variables. It assesses how well the relationship between two variables can be described using a monotonic function.

Rank correlation determines the degree of similarity between two sets of rankings assigned to a group of individuals or items. It is primarily used when data is qualitative (e.g., beauty, intelligence, honesty) or when the data contains extreme outliers that skew standard correlation calculations.

8. Define Relation between Correlation and regression

Correlation and regression are closely related statistical concepts. Correlation measures the degree and direction of relationship between two variables, while regression studies the functional relationship and helps in prediction.



The relationship between correlation coefficient (**r**) and regression coefficients (**b_{xy}** and **b_{yx}**) is:

$$r = \sqrt{b_{xy} \times b_{yx}}$$

Long question and answers

Q1. Define correlation and explain different types of correlation.

Answer. Correlation is a statistical technique used to measure the degree and direction of relationship between two variables. It shows whether the variables move together, move in opposite directions, or have no relationship at all.

01. Positive Correlation:

If both the variables vary in same direction is called positive correlation i.e., if one variable increases, the other also increases (or) if one variable decreases the other also decrease.

Ex:-

x	10	20	30	40	50
y	1	2	3	4	5

02. Negative Correlation :

If both the variables vary in opposite direction, i.e., called Negative Correlation i.e., if one variable increases, the other variable decreases (or) if one variable decreases, the other variable increases.

Ex:-

x	10	20	30	40	50
y	10	8	6	4	2



03. Simple Correlation :

When only two variables are studied i.e., the number of variables are given, choose relationship b/w only two variables.

04. Multiple Correlation :

When three (or) more variables are studied i.e., n variables are given, relationship b/w more than two variables.

05. Partial Correlation :

In this case, we study the relation between three (or) more variables and not all.

06. Total multiple Correlation:

In this case, we study the relationship between three (or) more variables it may be all also.

07. Linear Correlation:

If the amount of changes in one variable a constant ratio to the amount of changes in other variable then the correlation is said to be linear.

Ex:-

x	10	20	30	40	50
y	2	4	6	8	10

08. Non-Linear Correlation:

If the amount of change in one variable does not constant ratio to the amount of change the other variable then the correlation is said to be non-linear.

Ex:-



x	10	20	30	40	50
y	2	3	5	10	11

Q2. Find Karl Pearson's coefficient of correlation from the following data

Wages	100	101	102	102	100	99	97	98	96	95
Cost of living	98	99	99	97	95	92	95	94	90	91

Solution : Computation of Karl Pearson's coefficient of correlation

x	$X = x - \bar{x}, \bar{x} = 99$	X^2	y	$Y = y - \bar{y}, \bar{y} = 95$	Y^2	XY
100	+1	1	98	+3	9	+3
101	+2	4	99	+4	16	+8
102	+3	9	99	+4	16	+12
102	+3	9	97	+2	4	+6
100	+1	1	95	0	0	0
99	0	0	92	-3	9	0
97	-2	4	95	0	0	0
98	-1	1	94	-1	1	+1
96	-3	9	90	-5	25	+15
95	-4	16	91	-4	16	+16
990	0	54	950	0	96	61



$$r = \frac{\sum xy}{\sqrt{\sum x^2 \sum y^2}}, \quad r = \frac{61}{\sqrt{54 \times 96}} = 0.847$$

Q3.Following are the rank obtained by 10 students in two subjects, Statistics and Mathematics. To what extent the knowledge of the students in two subjects is related?

Statistics	1	2	3	4	5	6	7	8	9	10
Mathematics	2	4	1	5	3	9	7	10	6	8

Step-by-Step Solution

1. Calculate Differences D and Squares D^2

Student	Rank in Statistics x	Rank in Mathematics y	Difference $D = (x - y)$	D^2
1	1	2	-1	1
2	2	4	-2	4
3	3	1	+2	4
4	4	5	-1	1
5	5	3	+2	4
6	6	9	-3	9
7	7	7	0	0
8	8	10	-2	4
9	9	6	+3	9



10	10	8	+2	4
Total				$\sum D^2 = 40$

2. Apply Spearman's Rank Correlation Formula

The formula for Spearman's rank correlation coefficient r is:

$$R = 1 - \frac{6 \sum D^2}{n(n^2 - 1)}$$

$$R = 1 - \frac{6 \times 40}{10(10^2 - 1)}$$

$$R = 1 - \frac{240}{10(100 - 1)}$$

$$R = 1 - \frac{240}{10 \times 99}$$

$$R = 1 - \frac{240}{990}$$

$$R = 1 - \frac{24}{99}$$

$$R = 1 - 0.2424 = 0.7576$$

Q4. Determine the equation of a straight line which best fits the data.

X:	10	12	13	16	17	20	25
Y:	10	22	24	27	29	33	37

Solution : Straight line is $Y = a + bx$

Calculation Table

X	Y	X²	XY
10	10	100	100



12	22	144	264
13	24	169	312
16	27	256	432
17	29	289	493
20	33	400	660
25	37	625	925
$\Sigma X = 113$	$\Sigma Y = 182$	$\Sigma X^2 = 1983$	$\Sigma XY = 3186$

Number of data points (n) = 7

Normal Equations

The two normal equations for the straight line ($Y = a + bX$) are:

$$\Sigma Y = na + b\Sigma X$$

$$\Sigma XY = a\Sigma X + b\Sigma X^2$$

Substituting the summation values into the formulas:

$$182 = 7a + 113b \text{ (Equation 1)}$$

$$3186 = 113a + 1983b \text{ (Equation 2)}$$

Solving the Simultaneous Equations

To eliminate "a", multiply Equation (1) by 113 and Equation (2) by 7:

$$20574 = 791a + 12769b$$

$$22302 = 791a + 13881b$$

Subtract the first modified equation from the second:

$$1112b = 1728$$

$$b = 1728 / 1112$$

$$b \approx 1.56$$

Substitute the value of b back into Equation (1):

$$182 = 7a + 113(1.56)$$



$$182 = 7a + 176.28$$

$$7a = 5.72$$

$$a = 5.72 / 7$$

$$a \approx 0.82$$

$$Y = 0.82 + 1.56X$$

Q5. Calculate the regression equations of Y on X from the data given below, taking deviations from actual means of X and Y

Price (RS.)	10	12	13	12	16	15
Amount Demanded	40	38	43	45	37	43

Estimate the likely demand when the price is Rs. 20.

Solution : Calculation of Regression equation

x	(x – 13) = X	X ²	y	(y – 41) = Y	Y ²	XY
10	-3	9	40	-1	1	3
12	-1	1	38	-3	9	3
13	0	0	43	2	4	0
12	-1	1	45	4	16	-4
16	3	9	37	-4	16	-12
15	2	4	43	2	4	4

Regression equation of Y on X is:

$$\left((y - \bar{y}) = r \cdot \frac{\sigma_y}{\sigma_x} x - \bar{x} \right)$$
$$r \frac{\sigma_y}{\sigma_x} = \frac{\sum XY}{\sum X^2} = \frac{-6}{24} = -0.25$$



$$Y - 41 = -0.25(X - 13)$$

$$Y = -0.25X + 44.25$$

Q6.Difference between correlation and Regression

Correlation	Regression
Correlation measures the degree and direction of relationship between two variables.	Regression measures the effect of one variable on another and helps in prediction.
It shows whether variables move together.	It shows how much one variable changes when another variable changes.
There is no distinction between dependent and independent variables.	One variable is dependent and the other is independent.
Correlation coefficient is denoted by r .	Regression coefficients are denoted by b_{xy} and b_{yx} .
The value of r lies between -1 and +1 .	Regression coefficients may take any value.
It is mainly used to measure the strength of relationship.	It is mainly used for forecasting and estimation.
Only one correlation coefficient is obtained.	Two regression equations can be obtained: X on Y and Y on X.

7.Calculate Spearman's rank correlation coefficient for the data

X	48	33	40	9	16	16	65	24	16	57
Y	13	13	24	6	15	4	20	9	6	19

1. Ranking the Variables

Number of observations $N = 10$



For X: The value 16 is repeated **3 times**, so it shares the average of ranks 2, 3, and 4: $\frac{2+3+4}{3} = 3$.

For Y:

The value 6 is repeated **2 times** (sharing ranks 2 and 3): $\frac{2+3}{2} = 2.5$.

The value 13 is repeated **2 times** (sharing ranks 5 and 6): $\frac{5+6}{2} = 5.5$

2. Calculation Table

X	Rank x	Y	y	$D = R(x) - R(y)$	D^2
48	8	13	5.5	2.5	6.25
33	6	13	5.5	0.5	0.25
40	7	24	10	-3.0	9.00
9	1	6	2.5	-1.5	2.25
16	3	15	7	-4.0	16.00
16	3	4	1	2.0	4.00
65	10	20	9	1.0	1.00
24	5	9	4	1.0	1.00
16	3	6	2.5	0.5	0.25
57	9	19	8	1.0	1.00
Total					$\Sigma D^2 = 41$

3. Correction Factor for Ties

The correction formula for each tied group is $\frac{m(m^2-1)}{12}$, where M is the number of times a value repeats.

For X = 16 ($m_1 = 3$): $\frac{3(3^2-1)}{12} = \frac{24}{12} = 2$

For Y = 6 ($m_2 = 2$): $\frac{2(2^2-1)}{12} = \frac{6}{12} = 0.5$



For $Y = 13$ ($m_3 = 2$): $\frac{2(2^2-1)}{12} = \frac{6}{12} = 0.5$

$$\text{Total Correction} = 2 + 0.5 + 0.5 = 3$$

4. Applying the Rank Correlation Formula

$$r_s = 1 - \frac{6[\sum D^2 + \text{Correction Factors}]}{N(N^2 - 1)}$$

$$r_s = 1 - \frac{6[41 + 3]}{10(10^2 - 1)}$$

$$r_s = 1 - \frac{6 \times 44}{10 \times 99}$$

$$r_s = 1 - \frac{264}{990}$$

$$r_s = 1 - 0.2667 = \mathbf{0.7333}$$

The rank correlation coefficient is **0.7333**

8. From a sample of 200 pairs of observation the following quantities were calculated. $\sum X = 11.34$, $\sum Y = 20.78$, $\sum X^2 = 12.16$, $\sum Y^2 = 84.96$, $\sum XY = 22.13$. From the above data show how to compute the coefficients of the equation $Y = a + bX$.

Solution: We can compute the coefficients of the equation

$Y = a + bX$ by solving the normal equations

$$(\sum Y = na + b\sum X \text{ and } \sum XY = a\sum X + b\sum X^2)$$

Substituting the values

$$20.72 = 200a + 11.34b$$

$$a = \frac{20.72 - 11.34b}{200}$$

$$a = 0.1036 - 0.0567b$$

$$a = 0.1036 - 0.0567(1.82) \quad \text{and} \\ = 0.0005 \quad \therefore$$

$$22.13 = 11.34a + 12.16b$$

$$22.13 = 11.34(0.1036 - 0.0567b) + 12.16b$$

$$22.13 = 1.175 - 0.643b + 12.16b)$$



$$20.955 = 11.517 b$$

$$b = \frac{20.955}{11.517} = 1.82$$

$$Y = 0.0005 + 1.82X$$